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April 30th 2025

Multimodal Data Collection

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OVERVIEW

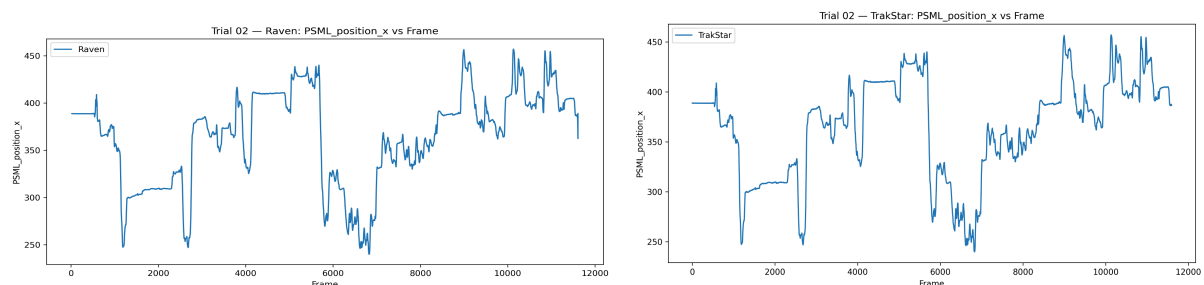
Task: Comparison of Kinematics from Raven Surgical Robot and TrakStar Motion Tracking System

Data Loading

I first mounted Google Drive so that Colab could see the raw files in my directory. That was done through `drive.mount('/content/drive', force_remount=True)`, which lets anyone re-run the cell without worrying about a stale mount from a previous session. All data lived in `/content/drive/MyDrive/January 2023`, and every peg-transfer trial resided in a folder whose name begins with the common stem `Peg_Transfer_SKay_T`. I used that piece of information to iterate through the folder and create a hashmap that contained the trial number as the key and value as the path to the folder in drive. This was done to allow scaling of code, meaning it could be used indefinitely while adding more and more data sets, just increasing the trial number. I then went through those folders to find the TrakStar and Raven csv files, once again taking advantage of the naming structure. Then, I set the number of rows as the minimum of the two files. And cut excess information from there.

Stacked Plots & Data Visualization

I then plotted all the columns against my frame column, excluding some of the irrelevant ones, such as 'True frame #', 'Trackstar time', etc... I did this for both the Raven and Trakstar csv files. I saved them into the two separate folders and did a visual analysis to see which variables corresponded to which. Example below (Rest are on Github):



It was kind of obvious just from glancing at the column names, but also just looking at the two, they look almost identical just shifted horizontally.

Table

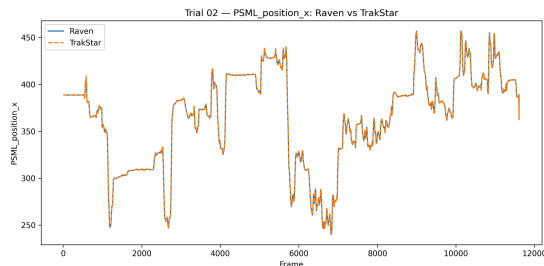
Proper Name:	Name in Raven	Name in TrakStar
PSML_position_x	PSML_position_x	PSML_position_x
PSML_position_y	PSML_position_y	PSML_position_y
PSML_position_z	PSML_position_z	PSML_position_z
PSMR_position_x	PSMR_position_x	PSMR_position_x
PSMR_position_y	PSMR_position_y	PSMR_position_y
PSMR_position_z	PSMR_position_z	PSMR_position_z
PSML_gripper_angle	PSML_gripper_angle	PSML_gripper_angle
PSMR_gripper_angle	PSMR_gripper_angle	PSMR_gripper_angle
Trackstar time	Trackstar time	Trackstar time
True frame #	True frame #	True frame #
Relative frame #	Frame	Relative frame #
Timestamp	Timestamp	Frame timestamp
PSML_velocity_x	PSML_velocity_x	N/A
PSML_velocity_y	PSML_velocity_y	N/A
PSML_velocity_z	PSML_velocity_z	N/A
PSMR_velocity_x	PSMR_velocity_x	N/A
PSMR_velocity_y	PSMR_velocity_y	N/A
PSMR_velocity_z	PSMR_velocity_z	N/A

Alignment Via Manipulation

I then had an idea, I compared their values at the same 'Timestamp' value. And the values were much closer, within $2 * 10^{-14}$ generally. This gave me the idea to align them, time-scale wise and number of data point wise. That was when I implemented a filter that removed all the rows that did not have time-values both files shared and then aligned them from that. This made my values align perfectly! Enough so that there was very little noise beyond a few decimal places down. I then went through and applied an offset just in case there was one, which didn't cause a noticeable improvement after my redesign of processing. Same with a scaling factor. This gave me a maximum error shown below:

 **1.1368683772161603e-13**

Overlaid Plots



There are some bumps, indicating a little bit of noise, but it is much better and seemingly in phase (Rest are on Github).

Bonus Task

I was unable to properly do the bonus, I tried accessing the video and did not have permissions and so I couldn't do it in time. I have quite a few finals coming up. My apologies.